AI LAB 8.py - C:/Users/K Aishwarya/Desktop/AI/AI LAB 8.py (3.14.2)

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```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

visited = []

def dfs(node):
    if node not in visited:
        visited.append(node)

        print("Visited Node:", node)

        for neighbour in graph[node]:
            dfs(neighbour)


print("DEPTH-FIRST SEARCH")
print("------------------")
print("Starting Node: A")
print("\nTraversal Order:")

dfs('A')

print("\nFinal DFS Traversal:", " -> ".join(visited))
```

```
>>>
============================================================
DEPTH-FIRST SEARCH
------------------
Starting Node: A

Traversal Order:
Visited Node: A
Visited Node: B
Visited Node: D
Visited Node: E
Visited Node: C
Visited Node: F
Visited Node: G

Final DFS Traversal: A -> B -> D -> E -> C -> F -> G
```